

# Mariam Ezzat MSc Student

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## Education

**MSc in Biomedical Engineering**, *University of Alberta*  
2025 - Present; **GPA:** 4.0

**Bachelor's Degree - Systems and Biomedical Engineering**,  
*Cairo University, Faculty of Engineering*  
2019 – 2024 | Giza, Egypt  
- **GPA:** 3.88/4; **IELTS:** 7.5/9.

## Experience & Internships

**Teaching Assistant (Part-time)**, *University of Alberta*  
July 2025 – April 2026

- Supporting instructors with Canvas setup.
- **Finite element method course:** grading assignments and exams.

**AI Career Accelerator Program**, *AMII*

**Teaching Assistant (Part-time)**, *Brock University*  
September 2024 – December 2024

- Graded assignments and exams for the **Cognitive Ergonomics** course.
- Taught **anatomy** labs, graded exams, and invigilated exams.

**Biomedical Intern**, *Global Medical City* 📍  
August 2023 – September 2023

- Performed daily check-ups on all the medical devices.
- Created an Error Log.
- Supervised the calibration and PM processes.

**Facilities and IT Intern**, *Integrated Diagnostics Holdings* 📍  
September 2022 – October 2022

- Acquired hands-on experience in Tendering and Medical Planning.
- Contributed to upgrading Web and Mobile App projects.

## Skills

### Programming Languages

Python, Java, Javascript, C/C++, SQL, Matlab, Bash.

### AI & Medical Imaging

PyTorch, TensorFlow, MONAI, OpenCV, Scikit-learn, Medical Image Analysis, Image Segmentation, 3D Surface Topography, DICOM, Data Annotation.

### Software & Deployment

Git, Docker, Flask, Django, Streamlit, REST APIs, HTML, CSS

## Publications & Conferences

**CNN-Based Deep Learning for Multi-Region Classification and Prediction of Cobb Angles from Markerless 3D Surface Topography in Adolescent Idiopathic Scoliosis**,  
*Alberta Biomedical Engineering Conference (2025)*

**Surface Topography-Based Analysis for Automated Classification of Adolescent Idiopathic Scoliosis (AIS)**  
*10th World Congress of Biomechanics (2026)*

**Reliability of an AI-Based Surface Topography Analysis for Scoliosis Screening and Cobb Angle Estimation: Preliminary Results**  
*Sosort International Congress (2026)*

**Facia-fix: mobile application for bell's palsy diagnosis and assessment using computer vision and deep learning** 📍,  
*Biomedical Physics & Engineering Express Journal (2024)*

## Technical Projects

**Multi-Region Cobb Angle Estimation via Deep Learning on 3D Surface Topography Scans**, *MSc Thesis Project*  
*Pytorch, DL, Open3D, Ensemble, Regression, Classification.*  
- Developed an ensemble deep learning framework for automated thoracic and lumbar Cobb angle estimation from 3D surface topography.

**Edge-Aware Multi-Class Vertebra Segmentation from Spinal X-ray Images**, *MSc Thesis Project*  
*DeepLabV3+, Edge Detection, Multi-Class Segmentation, Data Augmentation, Medical Imaging, PyTorch.*  
- Developed a vertebra segmentation model using DeepLabV3+ for multi-class spinal X-ray segmentation.

**AI-based Mobile Application for Bell's Palsy Diagnosis and Assessment** 📍, *Capstone Project*  
*DL, MediaPipe, Pytorch, Django, React, AWS, PostgreSQL.*  
- **Collaborator:** Uniparticle 📍.  
- **Cloud-based mobile** platform utilizing **deep learning** and **computer vision** for diagnosing and assessing facial paralysis, providing a **communication platform** between patient and doctor.

**TumorDetectAI: Brain Tumor Classification** 📍,  
*Python, EDA, DL, Sklearn, Tensorflow, DataProcessing*  
Brain Tumor classification using **deep learning**.

**SegmentaCV: Advanced Segmentation Toolkit** 📍,  
*Qt C++, Image Processing, OpenCV.*  
Desktop App **for thresholding and segmentation** algorithms from scratch.

**MammoDetect: Breast Cancer Classification** 📍,  
*Python, EDA, ML, Sklearn, Data Preprocessing.*  
Breast cancer classification using **machine learning**.

**ImageLab: Image Features Extraction And Matching** 📍,  
*QT C++, Image Processing, feature extraction and matching.*  
Desktop application that implements **different feature extraction and matching** algorithms.

**Face Recognition Web App** 📍,  
*Computer Vision, HTML, CSS, Javascript, Flask, ML, PCA.*  
Web app for face recognition using **Machine Learning** and **PCA**.

**VocalLock: Audio Fingerprint Recognition** 📍, *Data Acquisition, Feature Engineering, HTML, CSS, Javascript, ML, Flask.*  
Door Lock **web app** with **voice recognition** using **Machine Learning**.

## Extracurricular Activities

**Industry Advisory Council Administrator**, *University of Alberta*  
- **Ongoing:** Coordinate industry-academic meetings, track action items, and prepare meeting summaries

**Volunteer**, *BioAlberta Health and Life Science Showcase*  
- Assisted in organizing room panels.

**Registration Volunteer**, *Biomedical Engineering Research Day*  
- Managed participants' registration and on-site support.

**Head of Biomedical committee** 📍, *IEEE-CUSB*  
- Trained committee members to conduct workshops.  
- Coordinated curriculum development & delivery strategies.

**IT member**, *ASME-CUSS*  
- Designed and developed the front-end of the ASME club's website as an IT member, ensuring user-friendly navigation and aesthetic appeal.